

Before (size $\propto w_i$)**After (4 survive, 4 eliminated)** x_1  0.025 x_2  0.073 x_3  0.009 x_4  0.207 x_5  0.016 x_6  0.510 x_7  0.046 x_8  0.114draw
 $N = 8$  $\times 1$  $\times 2$  $\times 4$  $\times 1$

Systematic resampling: N pointers spaced $1/N$ apart land in segments of width w_i